STAT 4530/6530

Homework 2

Due: Monday, February 16 via eLC by 11:59 pm

Formatting instructions: Please label everything clearly and make sure that it is organized. I recommend typing your solutions, but neatly handwritten solutions are acceptable. For problems that require statistical computing, please attach relevant R code and plots. **Please write down (directly below your name) whether you are enrolled in STAT 4530 or STAT 6530**.

Grading: Homework accounts for 40% of your final grade, and all homework assignments contribute equally to your final homework score. Your score for each homework assignment will be recorded as a percentage of the total points that you received out of the total possible points. The amount of points possible for each problem are indicated in parentheses.

Problems

1. Consider the following study done at the National Institute of Science and Technology. Asbestos fibers on filters were counted as part of a project to develop measurement standards for asbestos concentration. Asbestos dissolved in water was spread on a filter, and 3-mm diameter punches were taken from the filter and mounted on a transmission electron microscope. An operator counted the number of fibers in each of 23 grid squares, yielding the following counts:

   31 29 19 18 31 28 34 27 34 30 16 18
   26 27 27 18 24 22 28 24 21 17 24

   We decide to model the counts as arising from a Poisson(μ) distribution. The probability mass function for this distribution is:

   $$f(y;\mu) = \frac{\mu^y e^{-\mu}}{y!}$$

   for $y = 0, 1, 2, \ldots$, and the distribution has mean and variance both equal to μ, where the rate parameter μ must be a positive real number.

   (a) (5 points) Find the maximum likelihood estimate of μ. Show your work (don't just write the answer, even though we did this in class).

   (b) (5 points) Find an approximate 90% confidence interval for μ.

Solution: Let $y_1, \ldots, y_{23}$ denote the observed counts. The log-likelihood is

$$\begin{aligned} l(\mu) &= \log\left[\prod_{i=1}^{23} \frac{\mu^{y_i} e^{-\mu}}{y_i!}\right] \\ &= \sum_{i=1}^{23} [y_i \log\mu - \mu - \log(y_i!)] \\ &= (\sum_{i=1}^{23} y_i)\log\mu - 23 * \mu - \sum_{i=1}^{23}\log(y_i!). \end{aligned}$$

We can differentiate the log-likelihood with respect to μ and set it equal to zero to find the critical point:

$$\frac{(\sum_{i=1}^{23} y_i)}{\mu} - 23 = 0.$$

This gives us the critical point $\hat{\mu} = \frac{1}{23}(\sum_{i=1}^{23} y_i$. If we plug in the actual values of the observations, we get $\hat{\mu} = 24.9$. To verify that this is the maximum likelihood estimate, we take the second derivative of the log-likelihood,

$$l''(\mu) = -\frac{(\sum_{i=1}^{23} y_i)}{\mu^2}$$

and note that it is negative for all μ. To find an approximate 90% confidence interval for μ, recall that for large n, $\hat{\mu}$ has an approximate $\mathrm{N}(\mu, 1/\mathcal{I}(\mu))$ distribution. Note that

$$\begin{aligned} \mathcal{I}(\mu) &= 23 * E\left[\left(\frac{\partial \log f(Y_1;\mu)}{\partial\mu}\right)^2\right] \\ &= 23 * E\left[\left(\frac{Y_1}{\mu} - 1\right)^2\right] \\ &= \frac{23}{\mu^2} * E\left[(Y_1 - \mu)^2\right] \\ &= \frac{23}{\mu^2} * Var\,(Y_1) \\ &= \frac{23\mu}{\mu^2} \\ &= \frac{23}{\mu}. \end{aligned}$$

Since we don't know μ, we can approximate $\mathcal{I}(\mu)$ by $\mathcal{I}(\hat{\mu}) = \frac{23}{24.9}$. Thus for large n, $\hat{\mu}$ has an approximate $\mathrm{N}(\mu, \frac{24.9}{23})$ distribution and an approximate 90% confidence interval for μ is

$$\hat{\mu} \pm z_{0.10/2}\sqrt{24.9/23}.$$

Since $z_{0.05} \approx 1.644854$, the approximate confidence interval is $(23.19, 26.61)$.

2. Consider an i.i.d. sample of size n from a distribution with probability mass function

$$f(y;p) = p(1-p)^{y-1}$$

for $y = 1, 2, 3, \ldots$, with $0 < p \leq 1$.

(a) (5 points) Find the maximum likelihood estimate of p.

(b) (5 points) Find the Fisher information $\mathcal{I}(p)$.

(c) (5 points) Find the asymptotic variance (meaning, the approximate variance when the sample size n is large) of the maximum likelihood estimate.

Solution: The log-likelihood is

$$\begin{aligned} l(p) &= \log\left[\prod_{i=1}^{n} p(1-p)^{y_i-1}\right] \\ &= \sum_{i=1}^{n}[y_i \log(1-p) - \log(1-p) + \log p] \\ &= (\sum_{i=1}^{n} y_i)\log(1-p) - n\log(1-p) + n\log p. \end{aligned}$$

We can differentiate the log-likelihood with respect to p and set it equal to zero to find the critical point:

$$-\frac{(\sum_{i=1}^{n} y_i)}{1-p} + \frac{n}{1-p} + \frac{n}{p} = 0,$$

which simplifies to

$$\frac{(\sum_{i=1}^{n} y_i)}{1-p} = \frac{n}{p(1-p)},$$

which yields critical point $\hat{p} = n/(\sum_{i=1}^{n} y_i) = 1/\overline{y}$. To verify that this is the maximum likelihood estimate, we take the second derivative of the log-likelihood,

$$\begin{aligned} l''(p) &= -\frac{(\sum_{i=1}^{n} y_i)}{(1-p)^2} + \frac{n}{(1-p)^2} - \frac{n}{p^2} \\ &= \frac{-(\sum_{i=1}^{n} y_i) + n}{(1-p)^2} - \frac{n}{p^2}, \end{aligned}$$

and note that it is negative for all possible values of p, since $-(\sum_{i=1}^{n} y_i) + n \leq 0$. Now let us compute $\mathcal{I}(p)$. First, note that

$$E(Y_1) = \frac{1}{p} \text{ and } Var(Y_1) = \frac{1-p}{p^2} \text{ and } E(Y_1^2) = Var(Y_1) + [E(Y_1)]^2 = \frac{2-p}{p^2}.$$

We have

$$
\begin{aligned}
\mathcal{I}(p) &= n * E\left[\left(\frac{\partial \log f(Y_1;p)}{\partial \mu}\right)^2\right] \\
&= n * E\left[\left(\frac{-Y_1)}{(1-p)^2} + \frac{1}{(1-p)^2} - \frac{1}{p^2}\right)^2\right] \\
&= n * E\left[\left(\frac{1-pY_1}{p(1-p)}\right)^2\right] \\
&= \frac{n}{p^2(1-p)^2} E\left[1 - 2pY_1 + p^2Y_1^2\right] \\
&= \frac{n}{p^2(1-p)^2}[1 - 2 + (2-p)] \\
&= \frac{n}{p^2(1-p)}.
\end{aligned}
$$

The asymptotic variance of the maximum likelihood estimate is

$$\frac{1}{\mathcal{I}(p)} = \frac{p^2(1-p)}{n}.$$

3. Consider an i.i.d. sample of size n from a $\mathrm{N}(\mu, \sigma^2)$ distribution.

   (a) (5 points) Show that the sample mean and sample variance make up a two-dimensional sufficient statistic for (μ, σ^2).

   (b) (5 points) Suppose we know that $\sigma^2 = 4$ but μ unknown. Find a sufficient statistic for μ.

   (c) (5 points) Suppose we know that $\mu = -3$ but σ^2 is unknown. Find a sufficient statistic for σ^2.

Solution: Let $\mathbf{y} = (y_1, \ldots, y_n)$ denote the sample. For part (a), the likelihood is

$$
\begin{aligned}
L(\mu, \sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(y_i - \mu)^2\right] \\
&= (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i-1}^{n}(y_i - \mu)^2\right] \\
&= (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\overline{y} - \mu)^2]\right].
\end{aligned}
$$

Here, we have used the fact that

$$\begin{aligned}
\sum_{i=1}^{n}(y_i-\mu)^2 &= \sum_{i=1}^{n}[(y_i-\overline{y})+(\overline{y}-\mu)]^2 \\
&= \sum_{i=1}^{n}\left[(y_i-\overline{y})^2+2(\overline{y}-\mu)(y_i-\overline{y})+(\overline{y}-\mu)^2\right] \\
&= \sum_{i=1}^{n}(y_i-\overline{y})^2+2(\overline{y}-\mu)\sum_{i=1}^{n}(y_i-\overline{y})+\sum_{i=1}^{n}(\overline{y}-\mu)^2 \\
&= (n-1)\frac{1}{n-1}\sum_{i=1}^{n}(y_i-\overline{y})^2+2(\overline{y}-\mu)(n\overline{y}-n\overline{y})+n(\overline{y}-\mu)^2 \\
&= (n-1)s^2+n(\overline{y}-\mu)^2.
\end{aligned}$$

Let $T(\mathbf{y})=(s^2,\overline{y})$, $h(\mathbf{y})=1$ and

$$g(T(\mathbf{y});\mu,\sigma^2)=(2\pi\sigma^2)^{-n/2}\exp\left[-\frac{1}{2\sigma^2}[(n-1)s^2+n(\overline{y}-\mu)^2]\right].$$

Then

$$L(\mu,\sigma^2)=g(T(\mathbf{y});\mu,\sigma^2)h(\mathbf{y}),$$

and it follows that $T(\mathbf{y})=(s^2,\overline{y})$ is sufficient for (μ,σ^2).

For part (b), the likelihood is

$$\begin{aligned}
L(\mu) &= \prod_{i=1}^{n}\frac{1}{\sqrt{8\pi}}\exp\left[-\frac{1}{8}(y_i-\mu)^2\right] \\
&= (8\pi)^{-n/2}\exp\left[-\frac{1}{8}\sum_{i-1}^{n}(y_i-\mu)^2\right] \\
&= (8\pi)^{-n/2}\exp\left[-\frac{1}{8}\left[\sum_{i=1}^{n}y_i^2-2\mu\sum_{i=1}^{n}y_i+n\mu^2\right]\right] \\
&= \exp\left[\frac{\mu}{4}\sum_{i=1}^{n}y_i-\frac{n\mu^2}{8},\right]*(8\pi)^{-n/2}\exp\left[-\frac{1}{8}\sum_{i=1}^{n}y_i^2\right]
\end{aligned}$$

If we let $T(\mathbf{y})=\sum_{i=1}^{n}y_i$,

$$h(\mathbf{y})=(8\pi)^{-n/2}\exp\left[-\frac{1}{8}\sum_{i=1}^{n}y_i^2\right],$$

and

$$g(T(\mathbf{y});\mu) = \exp\left[\frac{\mu}{4}\sum_{i=1}^{n} y_i - \frac{n\mu^2}{8},\right]$$

then

$$L(\mu) = g(T(\mathbf{y});\mu)h(\mathbf{y}),$$

and it follows that $T(\mathbf{y}) = \sum_{i=1}^{n} y_i$ is sufficient for μ.

For part (c), the likelihood is

$$\begin{aligned} L(\sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(y_i+3)^2\right] \\ &= (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i-1}^{n}(y_i+3)^2\right]. \end{aligned}$$

If we let $T(\mathbf{y}) = \sum_{i=1}^{n}(y_i+3)^2)$, $h(\mathbf{y}) = 1$ and

$$g(T(\mathbf{y});\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i-1}^{n}(y_i+3)^2\right],$$

then

$$L(\sigma^2) = g(T(\mathbf{y});\sigma^2)h(\mathbf{y}),$$

and it follows that $T(\mathbf{y}) = \sum_{i=1}^{n}(y_i+3)^2$ is sufficient for σ^2.

4. (10 points) A social scientist wanted to estimate the proportion of school children in Boston who live in a single-parent household. She decided to use a sample size such that, with probability 0.95, the error would not exceed 0.05. How large a sample size should she use, if she has no idea of the size of that proportion?

   Solution: Assume the samples are independent and each sample follows a Ber(p) distribution, where p represents the true proportion of school children who live in a single-parent household. Then if $\overline{Y}$ is the sample proportion of school children who live in a single-parent household, for a large sample size n, $\overline{Y}$ has an approximate normal distribution with mean p and standard error $\sqrt{p(1-p)/n}$. This implies that for any given sample, $\overline{y}$ will fall within 1.96 standard errors of p with probability 0.95. If the sample size n is such that the error is 0.05, we would have

$$0.05 = 1.96\sqrt{p(1-p)/n},$$

   and solving for n would give us

$$n = \frac{(1.96)^2 p(1-p)}{0.05^2}.$$

   We do not know p, but we know that $p(1-p)$ is maximized when $p = 0.50$. Since

$$\frac{(1.96)^2 0.50(0.50)}{0.05^2} = 384.16,$$

   taking $n = 385$ will ensure a margin of error that does not exceed 0.05.

5. Consider collecting a sample of size $n = 1497$ from a population with the goal of estimating the proportion of the population that prefers coffee rather than tea. Suppose that the true proportion is $p = 0.53$.

   (a) (10 points) Use R to to simulate s different samples of size n from the population (using the true value of p to simulate the data). For each sample, construct a 95% score confidence interval for p. Report the percentage of the s confidence intervals that contain the true value of p. Do this for $s = 5, 10, 100, 1000$.

   (b) (10 points) Repeat part [(a)], but this time construct 70% score confidence intervals.

   Solution: The code below computes the percentage of 95% confidence intervals that contain the true value of p for $s = 5$ and can be easily modified to handle the other cases.

```
s <- 5
n <- 1497
p <- 0.53
y <- rbinom(s, n, p)
conf.level = 0.95

score_conf_interval <- function(y, n, conf.level) {
  alpha <- 1 - conf.level
  z <- qnorm(1 - alpha / 2)

  p_hat <- y / n
  denominator <- 1+(z^2)/n

  center <- (p_hat + z^2/(2*n))/denominator
  half_width <- z*sqrt(
    (p_hat*(1-p_hat) + (z^2)/(4*n)) / n
  )/denominator

  lower_value <- max(0,center-half_width)
  upper_value <- min(1,center+half_width)

  return(c(lower = lower_value, upper = upper_value))
}

intervals <- sapply(y, score_conf_interval, n = n, conf.level=conf
   .level)

sum(intervals[1,] <= p & p <= intervals[2,])/s
```

   The results will vary a bit due to randomness of the samples, but for part (a) I got percentages 100, 100, 94, and 95.6 for $s = 5, 10, 100, 1000$, respectively. For part (b), I got percentages 80, 60, 65 and 70.4 for $s = 5, 10, 100, 1000$, respectively. The key point is that as s increases, the percentage of intervals that contain the true value of p get closer to the confidence level.

6. Consider a sample $\mathbf{Y} = (Y_1, \ldots, Y_n)$ that we wish to use to estimate the parameter θ. Suppose that $\theta_1(\mathbf{Y})$ is an estimator of θ with $E[(\theta_1(\mathbf{Y}))^2] < \infty$ for all θ, suppose that $T(\mathbf{Y})$ is a sufficient statistic for θ, and let $\theta_2(\mathbf{Y}) = E[\theta_1(\mathbf{Y})|T(\mathbf{Y})]$. Then, for all θ,

$$E[(\theta_2(\mathbf{Y}) - \theta)^2] \leq E[(\theta_1(\mathbf{Y}) - \theta)^2],$$

and the inequality is strict unless $\theta_1(\mathbf{Y}) = \theta_2(\mathbf{Y})$.

(a) (5 points) Note that $\theta_1(\mathbf{Y})$ and $\theta_2(\mathbf{Y})$ are both estimators of θ. Interpret the statement above in the context of comparing the two estimators and what it implies about using sufficient statistics to construct estimators.

(b) **This question is only required for students enrolled in STAT 6530.** (10 points) Provide a proof of the statement above. Hints: First, show that the two estimators have the same means, so it suffices to compare their variances. To compare the variances of the estimators, recall the following result: if X and Z are random variables and X has finite variance, then $Var(X) = E[Var(X|Z)] + Var(E[X|Z])$.

Solution: For [(a)], the statement provides some theoretical justification for the value of constructing estimators from sufficient statistics. In particular, if we start with an estimator that is *not* a function of a sufficient statistic, we can improve upon it by constructing an estimator that *is* a function of a sufficient statistic and is guaranteed to perform better in terms of mean squared error. For the proof required in [(b)], first note that we must compared the mean squared errors of the two estimators, but

$$E[\theta_2(\mathbf{Y})] = E[E[\theta_1(\mathbf{Y})|T(\mathbf{Y})]] = E[\theta_1(\mathbf{Y})].$$

Because the estimators have the same mean, it suffices to compare the variances. We have

$$\begin{aligned} Var(\theta_1(\mathbf{Y})) &= Var(E[\theta_1(\mathbf{Y})|T(\mathbf{Y})]) + E[Var(\theta_1(\mathbf{Y})|T(\mathbf{Y}))] \\ &= Var(\theta_2(\mathbf{Y})) + E[Var(\theta_1(\mathbf{Y})|T(\mathbf{Y}))] \\ &\geq Var(\theta_2(\mathbf{Y})). \end{aligned}$$

Equality holds only if $Var(\theta_1(\mathbf{Y})|T(\mathbf{Y})) = 0$, which is the case only if $\theta_1(\mathbf{Y})$ is a function of $T(\mathbf{Y})$. But if $\theta_1(\mathbf{Y})$ is a function of $T(\mathbf{Y})$, then $\theta_2(\mathbf{Y}) = E[\theta_1(\mathbf{Y})|T(\mathbf{Y})] = \theta_1(\mathbf{Y})$.